Noah E. Creany, PhD

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Research

Protected Area Conservation & Management

My expertise lies in Protected Area Conservation and Recreation Management, specifically applied, interdisciplinary research that integrates social science and ecology to inform management through a social-ecological systems lens.

- Recreation Ecology
- Human Dimensions of Wildland Recreation
- Visitor Use Management & Monitoring

Professional Experience

Postdoctoral Research Fellow 2024-2026

Oak Ridge Institute for Science and Education (ORISE)/ U.S. Forest Service

- Designed an algorithmic optimization approach for the U.S. Forest Service National Visitor Use Monitoring (NVUM) program to reduce program costs for data collection while maintaining the fidelity and utility of data.
- Developed gradient-boosted regression model using novel monitoring data to predict site-level daily forest visitation, and computer-vision object-detection workflow for automated trail-use monitoring.

Research Assistant 2017-2024

Utah State University

- Investigated recreation impacts on wildland ecosystems, employing advanced data analytics and machine learning techniques (Python, R) for regression, classification, and clustering.
- Managed and analyzed complex multi-modal datasets, including spatial, tabular, survey, and raster/imagery data.
- Collaborated with federal, state, and county land managers to ensure research deliverables met management priorities and timelines.

Education

Utah State University 2020-2024

PhD, Ecology *Logan, UT, USA*

- Research Assistant, Instructor of Record
- Dissertation: *Shifting Paradigms in Recreation and Parks and Protected Areas: Advancing Social-Ecological Systems Frameworks in Park and Protected Area Management*
- Major Professor: Dr. Christopher A. Monz, Supervising Committee: Dr. Mark Brunson, Dr. Wayne Freimund, Dr. Sarah Klain, Dr. Patrick Singleton.

Utah State University 2019
M.Sc., Recreation Resource Management *Logan, UT, USA*

- Thesis: *Kudos & KOMs: The Effect of Strava Use on Evaluations of Social and Managerial Conditions, Perceptions of Ecological Impacts, and Mountain Bike Spatial Behavior*
- Research Assistant/Teaching Assistant

Pennsylvania State University 2012
B.Sc., Community, Environment, and Development *University Park, PA, USA*

Funded Research

Arches National Park 2026
Transportation analysis of park entrance queue with computer vision - $5,000

City of Boulder Open Space and Mountain Parks (OSMP) 2026
Developing computer-vision visitor use monitoring methodology - $12,000

Rocky Mountain National Park Day Use Visitor Access Monitoring 2025
Rocky Mountain Conservancy $15,000

Rocky Mountain National Park Day Use Visitor Access Monitoring 2024
Rocky Mountain Conservancy $15,000

Awards

Utah State University Quinney College of Natural Resources 2023
Doctoral Student Researcher of the Year

Rocky Mountain Cooperative Ecosystem Studies Unit (CESU) 2023
Student Researcher (Honorable Mention)

Peer-Reviewed Articles

- **Creany, N.,** Monz, C., & Freimund, W. (2026). Strava Metro mobility data provides accurate estimates of recreation use in Urban-Proximate Protected Areas. *Journal of Outdoor Recreation and Tourism*. doi: 10.1016/j.jort.2025.101008
- Van Deursen, J., **Creany, N.,** Monz, C. A., & Freimund, W. (2025). Classifications of Recreation Specialization: Attitudinal and Behavioral Differences Across Specialization Types in the Nature Reserve of Orange County, CA, USA. *Journal of Park and Recreation Administration*. doi: 10.18666/JPRA-2025-12883
- Wheeler, I., Blair, P., **Creany, N.,** Manire, H., Morris, D., Mollett, S., Scearce, S. S., Stoellinger, T., Willoughby, J. R., & Dunning, K. H. (2025). Delisting the Northern Rocky Mountain gray wolf from the US Endangered Species Act: An assessment of political discourse over 20 years. *Wildlife Biology*. doi: 10.1002/wlb3.01589
- **Creany, N.,** Monz, C. A., & Esser, S. M. (2024). Understanding visitor attitudes towards the timed-entry reservation system in Rocky Mountain National Park: Contemporary managed access as a social-ecological system. *Journal of Outdoor Recreation and Tourism*. doi: 10.1016/j.jort.2024.100736

- Minehart, K., Antonio, A. D., **Creany, N.,** Monz, C., & Gutzwiller, K. (2024). Predicting trail condition using random forest models in urban-proximate nature reserves. *Environmental Challenges*. doi: 10.1016/j.envc.2024.100937
- Van Deursen, J., **Creany, N.,** Smith, B., Freimund, W., Avgar, T., & Monz, C. A. (2024). Recreation specialization: Resource selection functions as a predictive tool for protected area recreation management. *Applied Geography*. doi: 10.1016/j.apgeog.2024.103276
- Tomczyk, A. M., Ewertowski, M. W., **Creany, N.,** Ancin-Murguzur, F. J., & Monz, C. (2023). The application of unmanned aerial vehicle (UAV) surveys and GIS to the analysis and monitoring of recreational trail conditions. *International Journal of Applied Earth Observation and Geoinformation*. doi: 10.1016/j.jag.2023.103474
- **Creany, N.,** Monz, C. A., D'Antonio, A., Sisneros-Kidd, A., Wilkins, E. J., Nesbitt, J., & Mitrovich, M. (2021). Estimating trail use and visitor spatial distribution using mobile device data: An example from the nature reserve of orange county, California USA. *Environmental Challenges*. doi: 10.1016/j.envc.2021.100171
- Monz, C., **Creany, N.,** Nesbitt, J., & Mitrovich, M. (2021). Mobile Device Data Analysis to Determine the Demographics of Park Visitors. *Journal of Park and Recreation Administration*. doi: 10.18666/10.18666/JPRA-2020-10541
- Smith, J. W., Miller, A. B., Lamborn, C. C., Spernbauer, B. S., **Creany, N.,** Richards, J. C., Meyer, C., Nesbitt, J., Rempel, W., Wilkins, E. J., Miller, Z. D., Freimund, W., & Monz, C. (2021). Motivations and spatial behavior of OHV recreationists: A case-study from central Utah (USA). *Journal of Outdoor Recreation and Tourism*. doi: 10.1016/j.jort.2021.100426
- Wesstrom, S., **Creany, N.,** Monz, C. A., Miller, A., & D'Antonio, A. (2021). The Effect of a Vehicle Diversion Traffic Management Strategy on Spatio-Temporal Park Use: A Study in Rocky Mountain National Park, Colorado, Usa. *Journal of Park and Recreation Administration*. doi: 10.18666/jpra-2021-10746

Manuscripts in Preparation

- **Creany, N.**, White, E. - Network-based use-estimation for high visitation sites. Target Journal: *Journal of Forestry*.
- **Creany, N.**, Monz, C. - Direct Trail Management on High-Use Trails Improves the Quality of Recreation Experiences. Target Journal: *Landscape and Urban Planning*.

Natural Resource & Technical Reports

- Graham, R., Creany, N., & Monz, C. (2021). *Spatial Behavior of Backcountry Anglers and Hikers in Rocky Mountain National Park* [Natural Resource Technical Report]. National Park Service.
- Sisneros-Kidd, A., D'Antonio, A. L., Creany, N., Monz, C. A., & Shoenleber, C. (2019). *Recreation Use and Human Valuation on the Nature Reserve of Orange County, California*. Orange County, California: Utah State University.

Professional Affiliations

- Ecological Society of America (ESA) 2022-Present
- Recreation Ecology Research Network 2019-Present

Professional Service

Associate Editor 2026 - Present

Journal of Outdoor Recreation and Tourism

- Manage and referee the peer review process, ensuring high standards for research quality, robustness, and relevance.

Journal Peer Reviewer 2021-Present

Scientific Reports, Landscape and Urban Planning, J. of Park and Recreation Administration

- Peer-reviewed manuscripts related to park and protected area planning and management, the application of mobile device data for recreation use estimation, and recreation ecology.

Teaching Experience

Data Analysis and Programming for Natural Resource Management Spring 2024

Instructor of Record: NR6580 *Utah State University*

Wildland Recreation Behavior Fall 2024

Instructor of Record: ENVS 4500 *Utah State University*

Wildland Recreation Behavior Fall 2022

Instructor of Record: ENVS 4500 *Utah State University*

Wildland Recreation Behavior Fall 2021

Instructor of Record: ENVS 4500 *Utah State University*

Wildland Recreation Behavior Fall 2020

Instructor of Record: ENVS 4500 *Utah State University*

Technical Skills

Programming Languages

Python, R, ArcPy, SAS

Software

Google Cloud Compute, Google Earth Engine, ArcGIS, QGIS, Adobe CC, LaTex, Quarto